ALTair

Neural Architecture & Reproducible Software Baseline

Team Work Split — 3 People • Rover mobility anomaly detection • GF180 research ASIC

WHAT THIS DOCUMENT IS

The ALTair spec defines 6 owner roles. This splits them into **3 balanced parts** that follow the project's natural order: get legitimate data and prove simple methods first, then build and freeze the neural network in exact math, then wrap it in firmware and size it for the chip. Project-lead duties (scope, decisions, final signatures) are shared and held by the team lead on top of their part.

THREE RULES EVERY PERSON MUST FOLLOW

1. **Scientific honesty** — only claim what the data actually proves; if the fancy SNN isn't better than a simple method, say NO-GO. 2. **No data leakage** — test data must never touch training, splits are by whole drive/session. 3. **Bit-exact** — the final integer math must match perfectly (Python = C = RTL), zero mismatches. Everything is versioned with checksums and git commits.

Person 1 — Data & ML Baselines (Phases P0–P2 • Gates G0, G1, G2, G3-data)

Owns the foundation: get legitimately-accessible data, prove there's no leakage, and build every simple "could a basic method do this?" baseline before anyone trusts a neural network. If the data isn't clean and legal, nothing downstream is valid — so this part gates the whole project.

Tasks

ID	Task	What it includes
A1	Paper & data audit (P0)	Read the reference paper closely; write down raw vs derived signals, the 8 Hz rate, 4s/1s windowing, features, and the normal/anomalous definitions, with citations.
A2	Data access & legal check (P0)	Verify what data is actually downloadable and licensed (RoSE rover set, NASA resources). Mark anything unavailable as BLOCKED. No proprietary data in the repo.
A3	Dataset card + manifests (P0/P1)	Document sources, units, sampling rates, missing data, provenance and SHA-256 checksums. Parse and sanity-check the first few trials.
A4	Task charter & label ontology (P1)	Decide clearly: off-nominal mobility detection (Task A) vs real motor-fault detection (Task B). Only claim Task B if valid fault labels exist. Define NOMINAL / TERRAIN-SLIP / CALIBRATION / CONTROLLED_FAULT / UNKNOWN.
A5	Group-safe splits + leakage tests (P1)	Split by whole drives/sessions FIRST, then make windows. Keep one episode in one split. Automated checks for duplicate files, timestamp overlaps, contiguous-window leakage. Fit scaling only on training.
A6	Baseline ladder B0–B3 (P2)	B0 simple rules (current/speed thresholds), B1 classical (z-score, PCA, Isolation Forest), B2 lightweight ANN (small MLP/GRU/autoencoder), B3 CAIDDA-style autoencoder (only if source features exist).
A7	Run matrix + metrics framework (P2)	Run all baselines on identical grouped splits, ≥3 seeds, ≥2 window sizes, ≥2 false-alarm operating points. Report AUROC/AUPRC, false-alarms per hour, with confidence intervals and negative results.

Deliverables

File	Acceptance
docs/paper_audit.md	Parameter extraction with source-section references.
docs/data_card.md + data/manifests/*.json	Rights, sources, units, missingness, provenance, checksums.
data/splits/frozen_split.json	Session-disjoint partitions, zero leakage (automated proof).
experiments/index.csv + reports/	Complete B0–B3 comparison with uncertainties and negative findings.

Person 2 — SNN & Fixed-Point Golden Model (Phases P3–P4 • Gates G3, G4)

Owns the actual neural network and the exact math behind it. Builds and fairly sweeps the spiking network (SNN), then turns the chosen model into integer-only math that a chip can run, and proves the Python and C versions match bit-for-bit. This is the most technically deep part.

Tasks

ID	Task	What it includes
B1	SNN candidate families (P3)	Try feed-forward LIF, recurrent/temporal LIF, spiking reconstruction, and hybrid fixed-features+SNN. Nothing frozen yet; 256-neuron RTL is only a capacity reference, don't force the model into it.
B2	Mandatory parameter sweep (P3)	≥2 spike encoders (delta/event + rate/level), simplest realistic LIF neuron, ≥3 timestep counts T, ≥3 topology sizes. Log energy proxy (spikes) and detection latency.
B3	Ablations & negative results (P3)	Shuffled-label / no-timing controls (prove it learns real structure, not leakage); current-only vs +speed vs +IMU; with/without spike encoding. If SNN loses to a simpler method, recommend NO-GO.
B4	Quantization contract (P4)	Freeze exact arithmetic: signedness, Q-format, scale/offset, clipping; weight bit-width & layout; neuron-state format, leak, threshold, reset rule, saturation; accumulator width proof; exact loop order; anomaly-score math and ≥/> comparison.
B5	Integer + C golden model (P4)	Build: (1) float research model, (2) deterministic quantized Python model, (3) integer-only golden model, (4) minimal C reference using only fixed-width integers, (5) vector-by-vector Python-vs-C comparison.
B6	Golden-vector package (P4)	Normal, off-nominal, adversarial and random seeded vectors. For each: raw packet, preprocessed values, encoded spikes, per-timestep neuron states, output, anomaly score, decision, cycle budget and hash. JSON/CSV + binary. **Zero mismatches** required (Gate G4).

Deliverables

File	Acceptance
docs/model_card.md	Task, limits, architecture, training, metrics, calibration, dataset shift.
export/model_package/**	Weights, state formats, thresholds, normalizers, exact shapes, hashes.
export/golden_vectors/**	Boundary + normal traces; integer Python and C are bit-identical.

Person 3 — Firmware, Hardware Sizing & Verification (Phases P5–P6 • Gates G5, G6, verification for G4/G6)

Owns everything that turns the model into a real, runnable, chip-ready system: the firmware that feeds sensor data in and reads results out, the memory/speed/area numbers the chip team needs, and the independent checking that nobody faked their results.

Tasks

ID	Task	What it includes
C1	Telemetry input contract (P5/FW1)	Define signal IDs, calibrated units, sampling rates, timestamps, and how missing/out-of-order data is handled. Validate schema roundtrip.
C2	Streaming & buffer owner (P5/FW2)	Circular/double buffer, window overlap, IRQ/DMA ownership, overflow recovery. Prove no data races under stress and no missed periodic windows (32 samples per 4s window).
C3	Reference preprocessing + model loader (P5/FW3-FW4)	Preprocessing library that is bit-equal to the integer model; model loader with checksum/version check that rejects corrupted or mismatched weights.
C4	Driver, register map & result packet (P5/FW5-FW6)	Symbolic register map (CONFIG, CONTROL, STATUS, INPUT_BASE, MODEL_ID, OUTPUT_SCORE, IRQ_STATUS, ERROR_CODE — no invented addresses). Driver states RESET→CONFIGURED→BUFFER_READY→BUSY→RESULT_READY/ERROR. Versioned result packet + error reasons.
C5	PC replay + bring-up self-tests (P5/FW7-FW8)	Replay a fixed dataset stream at/above required rate (HIL); power-on and on-request deterministic self-tests.
C6	Independent verification (G4/G6)	Replay Person 2's golden vectors independently; adversarial input tests; reproducible

		regression; trace comparisons. Confirm bit-true results on a clean second environment.
C7	Chip sizing sheet + architecture handoff (P6)	Compute weights bytes, live neuron-state bytes, input-buffer bytes, total memory; per-timestep operations; worst-case cycles; throughput vs stride; I/O bandwidth. Compare options (Ibex-only vs SNN accelerator vs keep/remove NPU). Produce ADR-FINAL with the alternatives matrix.

Deliverables

File	Acceptance
docs/interface_contract.md	Timing, signal schema, registers/API, packets, interrupts, recovery.
src/firmware_ref/** + tests/replay/**	Deterministic replay and streaming stress tests pass.
docs/architecture_handoff.md	Memory table, cycles, bandwidth, option matrix, open dependencies.
decisions/ADR-FINAL.md + freeze_signoff.md	Trade-offs, GO/NO-GO evidence, all four signatures.

How the three parts connect

The work is a pipeline — each person hands the next a frozen, checksummed result. Nothing proceeds on a guess.

Handoff	From → To	What must be agreed
Data → Model	Person 1 → Person 2	Frozen splits, the deployable signal list, and which task (A or B) is valid. Person 2 trains only on this.
Model → Firmware	Person 2 → Person 3	Exact quantization contract + golden vectors. Person 3's preprocessing must be bit-equal to the integer model.
Verify (loops back)	Person 3 → Person 2	Independent replay of golden vectors. Any mismatch = model is NOT hardware-ready; Person 2 fixes before freeze.
Sizing → chip team	Person 3 → RTL/ASIC	Memory/cycle/area numbers confirmed against real GF180 macros before any freeze.

Gates (nobody skips these)

Gate	Must pass	Owner
G0 / G1	Data is legal & sourced; a real labelled task exists (else only off-nominal detection).	Person 1
G2	Clean data contract — frozen splits, no leakage.	Person 1
G3	Model candidate justified — SNN not worse than a matched simple baseline.	Person 1 + Person 2
G4	Bit-true — zero mismatches across all golden vectors.	Person 2 + Person 3 (verify)
G5	Firmware contract — streaming, driver, recovery signed off.	Person 3
G6	Architecture freeze — memory/cycle/IO fit real hardware; full evidence pack.	Person 3 + leads

SUGGESTED START ORDER

Person 1 starts immediately (P0 data audit — the critical path; nothing is valid until data is legal and clean). Person 2 can begin SNN research on public/sample data in parallel, then lock onto Person 1's frozen splits once ready. Person 3 can draft the firmware interface and sizing spreadsheet early using symbolic values, then fill real numbers as the model freezes. The schedule in the spec (~45 days) is indicative only — data access and board availability are the real bottlenecks.